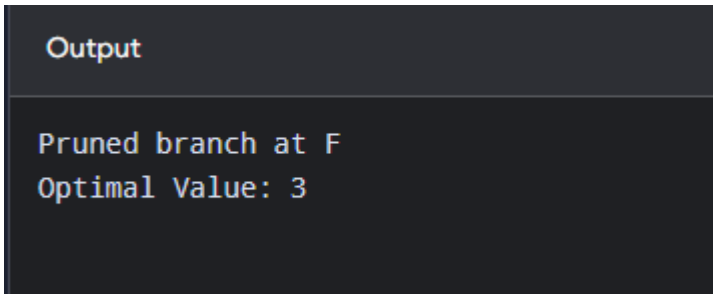
Lab experiment

Alpha-Beta Pruning Algorithm for Gaming

Coding:

```
# Alpha-Beta Pruning Algorithm for Gaming

# Game tree
tree = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G']
}

# Utility values of leaf nodes
values = {
    'D': 3,
    'E': 5,
    'F': 2,
    'G': 9
}

# Alpha-Beta Pruning function
def alpha_beta(node, alpha, beta, is_max):
    if node not in tree:
        return values[node]

    if is_max:
        best = float('-inf')

        for child in tree[node]:
            value = alpha_beta(child, alpha, beta, False)
            best = max(best, value)

            alpha = max(alpha, best)

            if beta <= alpha:
                print("Pruned branch at", child)
                break

        return best
```

```
    else:
        best = float('inf')

        for child in tree[node]:
            value = alpha_beta(child, alpha, beta, True)
            best = min(best, value)

            beta = min(beta, best)

            if beta <= alpha:
                print("Pruned branch at", child)
                break

        return best

# Main program
result = alpha_beta(
    'A',
    float('-inf'),
    float('inf'),
    True
)

print("Optimal Value:", result)
```

Output: